Multimodal Classification of Mobile EEG and Eye-Tracking During Outdoor Walking

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(Dated: June 6, 2026)

Brain Computer Interfaces (BCI) typically use electroencephalography (EEG) to interface with a computer via brain signals. However, EEG is susceptible to motion artifacts, including muscle activity, resulting in a large amount of noise. This is especially apparent during physical activity like walking. In this paper, we present multimodal methods of classification for a mobile EEG and eye-tracking (ET) setup during real-world walking, coupling the EEG signal with gaze information collected from eye-tracking. We demonstrate classifier performance on a dataset that tests inhibitory control across four conditions (sit/walk × attend/unattend). We evaluated three deep-learning classifiers, EEGNet (EEG only), RawGazeFusionNet (EEG + raw ET), and SemanticGazeFusionNet (EEG + semantic gaze labels via ResNet-50), alongside traditional ML baselines using pre-specfied spectral features (LDA, SVM, logistic regression). The models were applied to two classification problems: Go vs. NoGo (hits vs. correct rejections) and within the NoGo trails (correct rejection vs. commission errors). We found that EEGNet outperforms the traditional machine learning baselines; however, multimodal fusion did not significantly improve over EEG alone. Additionally, walking degrades classifier performance by roughly 4–9 percentage points relative to a stationary lab setting, and walk-trained models transfer better to sitting than vice versa.

I. INTRODUCTION

The Brain Computer Interface (BCI) technology allows for the control of a computer through brain signals [Wolpaw et al., 2002]. Although BCIs have traditionally been tied to assistive technologies [Schwartz et al., 2006], some of their most compelling applications involve healthy users who must sustain performance during continuous, real-world tasks: an athlete monitoring focus during training, a driver whose vehicular interface adapts when their attention lapses, or a soldier maintaining situational awareness under load, where a lapse can be life-or-death. There is also growing interest in non-invasive BCIs for everyday computing, including consumer EEG headsets such as Emotiv or Muse and the integration of EEG with VR; more broadly, the field is moving toward ubiquitous BCI, where computing can happen anywhere. Yet most BCIs are based on lab studies with ideal conditions and rarely replicate real-world settings [Ding et al., 2019]. Mobile EEG work has shown that it is possible to record meaningful brain signals outside the lab, and that ERP signals can survive walking in the real world, although classification under these conditions remains difficult [Debener et al., 2012; De Sanctis et al., 2014; Reiser et al., 2025]. In this work, we ask how well deep-learning BCI classifiers perform on mobile EEG recorded during real-world walking, and whether fusing gaze information improves them.

Most BCIs, however, are validated in controlled, stationary lab settings, whereas the real-world scenarios above all involve movement, noise, and divided attention. For these use cases EEG is a powerful tool, because it has high temporal resolution and is non-invasive. However, EEG is notoriously susceptible to motion artifacts, including blinks and muscle movements, especially during physical activity. Examining how EEG BCIs behave during walking therefore reveals how these systems react to real-world artifacts, and points toward potential solutions.

Traditional ERP (event-related potential) EEG classification has relied on spatial filtering approaches like xDAWN and Riemannian Geometry [Lotte et al., 2018], however certain deep learning methods like CNNs or more specifically EEGNet [Lawhern et al., 2018] have emerged as a strong alternative, allowing for less manual feature engineering. A key limitation of unimodal EEG approaches is that performance degrades under real-world conditions such as walking, as artifacts may contaminate and add noise to the EEG signal [Ding et al., 2019]. To address this limitation, multimodal approaches fuse EEG with additional signals; prior work has demonstrated the efficacy of this strategy using non-scalp electrodes such as horizontal and vertical EOG around the eye and EMG on the upper back [Ding et al., 2019]. Eye tracking is a particularly attractive complementary modality in this context: fusing EEG with gaze or eye-movement signals has been shown to improve classification of attention and related mental states relative to EEG alone [Cheng et al., 2020; Vortmann et al., 2022].

In this paper, we take advantage of an ambulatory EEG dataset that was recently collected at UCSB. In this study, participants' physical state (sitting, walking) and auditory attention (attend, unattend) were manipulated while EEG and eye-tracking data were recorded. Here, we use the dataset to test how well a deep learning BCI classifier performs under various conditions. Prior to feeding in the gaze data as a modality into the classifier, the pipeline includes a computer vision component that classifies what the participant was looking at from the world video. To help address the increased noise in EEG during physical activity, we add gaze as a comple-

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mentary modality alongside EEG. We hypothesize that (i) gaze fusion will improve classification over EEG alone, (ii) walking will degrade performance relative to sitting, consistent with mobile-EEG studies reporting attenuated ERPs and lower classification accuracy during locomotion [Zink et al., 2016], and (iii) cross-condition transfer will be asymmetric, with models trained on the walking condition transferring better to sitting than the reverse, motivated by the principle that training on noisier, more variable data can act as a regularizer that improves robustness and generalization [Bishop, 1995].

II. RELATED WORK

A. EEG

EEG is a method for noninvasive measurement of brain activity from electrodes placed on the scalp [Berger, 1929]. The measured signal reflects post-synaptic electrical activity from large, synchronously active populations of neurons, providing a direct measure of neural activity with high temporal resolution [Nunez & Srinivasan, 2006]. To isolate the voltage associated with a specific task event, EEG studies use the event-related potential (ERP) technique, in which the signal is time-locked to an event and averaged across many repetitions so that random background activity cancels out, leaving a waveform that reflects the neural response to that event [Luck, 2014]. This study focuses on two ERP components. The P3 (or P300) is a positive deflection peaking roughly 300–600 ms after an infrequent or task-relevant stimulus and is a canonical index of attention and stimulus evaluation [Polich, 2007; Benikos et al., 2013]. The N2 (or N200) is a negative deflection peaking around 200–350 ms; its frontocentral variant is associated with cognitive control, in particular conflict monitoring and response inhibition, which makes it especially relevant for go/no-go tasks [Folstein & Van Petten, 2007].

Beyond time-domain ERPs, the EEG signal can be decomposed into oscillatory power across frequency bands, several of which are informative for attention and inhibitory control. Posterior alpha ($\sim$8–12 Hz) is a well-established marker of covert spatial attention: alpha power lateralizes across parieto-occipital cortex with the locus of attention and is thought to gate or suppress task-irrelevant input [Thut et al., 2006; Foster & Awh, 2019]. Frontal-midline theta ($\sim$4–8 Hz) increases with cognitive-control demands such as conflict monitoring and response inhibition [Cavanagh & Frank, 2014], while sensorimotor beta ($\sim$13–30 Hz) is linked to motor preparation and to the cancellation or withholding of movement [Engel & Fries, 2010]. Together, these bands motivate the spectral features (frontal theta, central beta, and frontal/parietal/occipital alpha power) extracted for our scalar baselines.

To translate these neural features into predictions, a range of classifiers has been applied to EEG. Classical pipelines typically combine spatial filtering, such as xDAWN or common spatial patterns, or covariance-based Riemannian-geometry representations, with linear classifiers such as LDA, support vector machines, or logistic regression [Lotte et al., 2018]. These approaches are interpretable and can work with limited data, but they depend on hand-crafted features and often transfer poorly across subjects and recording conditions. More recently, deep learning has offered an end-to-end alternative that learns features directly from the signal. EEGNet [Lawhern et al., 2018] is a representative example: a compact convolutional network that uses depthwise and separable convolutions to capture temporal and spatial structure with relatively few parameters, achieving performance comparable to state-of-the-art methods across multiple BCI paradigms while remaining lightweight enough to train on limited data. Accordingly, we use the classical feature-based models as interpretable baselines and EEGNet as the deep-learning backbone in this work.

B. Ambulatory EEG/BCIs

Even within BCIs, P300-based BCIs are among the most well-studied paradigms [Lawhern et al., 2018; Ding et al., 2019]. A key advantage of these technologies is that they open the door to ubiquitous BCI, where computing occurs at any time or place. In such a paradigm, users may interact with the BCI system while moving around their environment or engaging in physical activity. However, EEG is particularly sensitive to biophysiological artifacts associated with movement [Ding et al., 2019; Debener et al., 2012]. To ensure that these systems remain functional, it is important to understand how EEG classifiers behave under real-world conditions. A further challenge is that, for BCI paradigms in particular, models often struggle with cross-subject generalization owing to individual differences [Ding et al., 2019]. The present study does not aim to build a real-time BCI; rather, we analyze the data offline to assess how a potential future BCI might perform when faced with different recording conditions, including physical movement.

One of the challenges with a nonstationary EEG setup is that signal characteristics shift with fatigue, arousal and physical state [Ding et al., 2019]. In the outdoor paradigm, mobile EEG ERPs have been shown to survive motion artifacts to a meaningful degree [Debener et al., 2012; Nathan & Contreras-Vidal, 2016]. In particular, walking imposes cognitive demands that compete with concurrent cognitive tasks such as an attention task [Pizzamiglio et al., 2017; Patelaki et al., 2022; Peskar et al., 2023]. More specifically, for EEG signals during walking, the alpha signal decreases, artifacts increase and ERPs are attenuated, but often detectable [Jacobsen et al., 2020; Reiser et al., 2025; Ladouce et al., 2017]. Additionally, walking recruits different visual processing patterns, suggesting that spatial attention is not just noisier or reduced, but rather reorganized and redistributed in ways that differ from being stationary [Cao and Händel., 2019; Davidson et al., 2024].

III. DATASET

The present dataset manipulates physical state and attention through two means: attend versus unattend, and sitting versus walking. The auditory go/no-go task serves as the paradigm for studying inhibitory control. While the parietal P3 reflects target/oddball processing, the frontocentral NoGo-P3 is associated with successful response inhibition, making correct rejection vs. failed inhibition the primary BCI classification target [Benikos et al., 2013; Johnson., 1993]. The dataset includes the following four conditions: sit-attend, sit-unattend, walk-attend, and walk-unattend. Figure 1 shows the task structure (panel A) together with the outdoor walking route and environment (panels B and C).

Twenty adult volunteers from the University of California, Santa Barbara community participated in the study and received financial compensation of $20 per hour.

Auditory stimuli consisted of three tones presented through over-ear headphones at frequencies of 500 Hz, 1000 Hz, and 1500 Hz, all delivered at a fixed inter-stimulus interval (ISI) of 1,200 ms. The experiment employed an inhibitory control paradigm in which participants monitored a continuous stream of tones and were instructed to respond with a button press to every tone unless the current tone was a direct repetition of the immediately preceding tone (target probability $p = 0.15$). Upon detecting a repeated tone, participants were required to withhold their response, making successful withholding the correct rejection trials of interest for the BCI classification problem. Responses were recorded via a wireless remote mouse. Tone presentation order was pseudorandomized across blocks to prevent participants from anticipating predictable sequences.

The task was organized into five blocks per condition, with each block comprising 200 tones and short self-paced rest breaks permitted between blocks, yielding approximately 20 minutes of stimulus presentation per condition. Prior to the main experiment, each participant was fitted with the EEG cap and completed approximately 100 practice trials while seated in the laboratory to ensure task comprehension and stable performance before data collection began.

Each participant completed four counterbalanced experimental conditions in a single session: attend-sit, attend-walk, unattend-sit, and unattend-walk. In the sitting conditions, participants were seated inside the laboratory facing a plain white wall and instructed to minimize movement. In the walking conditions, participants walked an approximately one-mile loop around the UCSB lagoon along a path consisting of varied terrain including dirt, broken asphalt, and sand. The route exposed participants to naturalistic visual distractors including ocean and lagoon views, surrounding foliage, and unpredictable environmental events such as pedestrians, cyclists, wildlife, and maintenance vehicles. Participants were briefed on the designated route before each walk, and an experimenter followed at a discreet distance to provide navigational assistance if needed.

EEG data was recorded continuously using a mobile BrainProducts LiveAmp system (Brain Products GmbH, Munich, Germany).Thirty-two active electrodes were mounted in an elastic cap and positioned according to the extended International 10-20 system, with two additional electrodes placed at the left and right mastoids for referencing. Data were sampled at 500 Hz and referenced offline to the average mastoid signal.

Eye tracking data were recorded concurrently using Pupil Labs Neon glasses (Pupil Labs GmbH, Berlin, Germany) paired with a Motorola mobile phone for data logging. The glasses captured continuous gaze data including all fixation events throughout the task. Triggers from both the stimulus presentation software and the participant response device were embedded in the eye tracking recording stream, enabling time-synchronized alignment of fixation data with stimulus and response events.

To characterize participant visual behavior during the walking conditions, frames from the behavioral video recordings were reviewed and labeled by five independent human annotators, who produced a total of 8,570 behavioral labels across all participants and conditions. Fixation categories included trail/ground, vegetation, sky, water, ocean, people, and other; water and ocean were subsequently collapsed into a single category for analysis. Inter-rater reliability was assessed using Fleiss's Kappa, yielding a value of 0.953, indicating strong consistency across labelers.

The final dataset comprised 20 subjects across 4 conditions with approximately 1,000 trials per condition organized into 5 behavioral blocks, yielding a large trial pool for model training and evaluation. However, the dataset presents three notable challenges. First, the fixation label distribution is severely imbalanced, with trail/ground accounting for approximately 76% of all labeled fixations, which motivated the use of focal loss during model training. Second, the number of usable EEG NoGo trials (cases where the tone is repeated and participants are expected to withhold their response) is quite limited (13 subjects had fewer than about 50 failed-inhibition trials), which is especially problematic when comparing correct inhibition against failed inhibition. Deep-learning paradigms typically require about 50–100 trials per class. Third, the EEG data exhibit substantial inter-subject variability and motion-related noise, both of which required extensive preprocessing and motivated the LOSO cross-validation scheme as a test of cross-subject generalization.

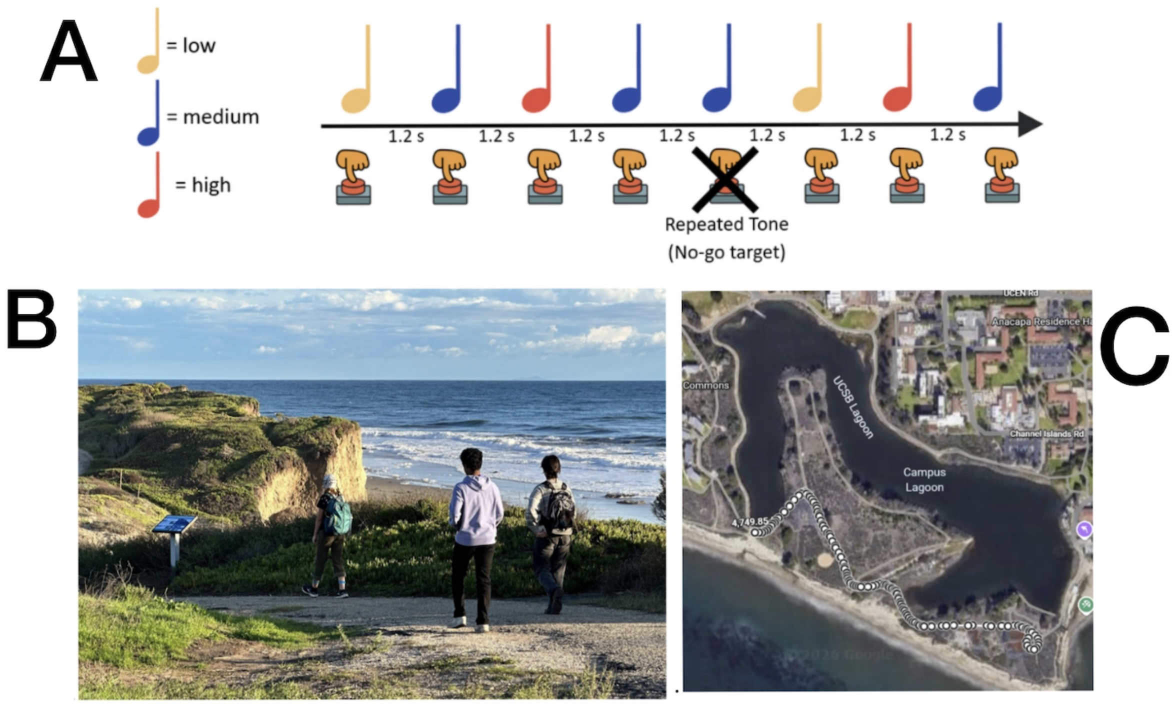

FIG. 1. Dataset task and walking environment. A: detailed auditory Go/NoGo task structure. B: representative image of the outdoor walking course. C: map of the UCSB lagoon route used during mobile EEG and eye-tracking collection.

IV. METHODS

The present study compared how EEGNet (unimodal), RawGazeFusionNet (EEG + raw ET time series), and SemanticGazeFusionNet (EEG + labeled gaze data) perform when classifying correct versus failed inhibition and Go versus NoGo trials, with non-deep-learning methods as a baseline comparison. Figure 2 summarizes the full analysis pipeline. Because participants do not respond during the unattend conditions, those conditions are excluded from the classifiers and used only as a sanity check that EEG preprocessing produces the expected ERPs (P3). In addition, because there are no scenes to classify in the sitting conditions, the gaze-fusion models are applied only to the walking conditions.

A. EEG preprocessing

This current experiment employs the following preprocessing chain: BrainVision 32-channel cap, downsampled to 250 Hz, post-hoc fixes of the data includes trigger latency and montage correction, next ICA is applied for artifact removal and the data is epoched from -200 ms to 800 ms. The EEG event codes are also aligned with the behavioral data. Figure 3 shows the P3 ERPs at Pz for Go vs. NoGo trials.

B. ET preprocessing

Eye-tracking data were exported from the Pupil Labs Neon system and processed per subject and condition. For each trial, event trigger codes were parsed from the annotations file and used to define a time-locked window around stimulus onset. Within each window, raw gaze samples were extracted from the continuous gaze position stream and used to compute mean horizontal and vertical gaze position (in pixels) as well as mean pupil diameter (in mm). Where pupil diameter was absent or unreliable in the primary gaze file, values were drawn from the 3D eye states file by averaging available left and right pupil measurements across the trial window. Each trial thus yielded a single row of ET features (gaze position and pupil size) aligned to the corresponding EEG epoch by shared trigger timestamp.

C. Computer Vision Methods

The world camera video was temporally aligned with the gaze data. Each gaze fixation was matched with the corresponding frame from the world video, and a 224 $\times$ 224 pixel region was cropped around the gaze position for that fixation. A subset of crops (n = 8,570 out of N = 131,511 total crops across subjects and conditions)

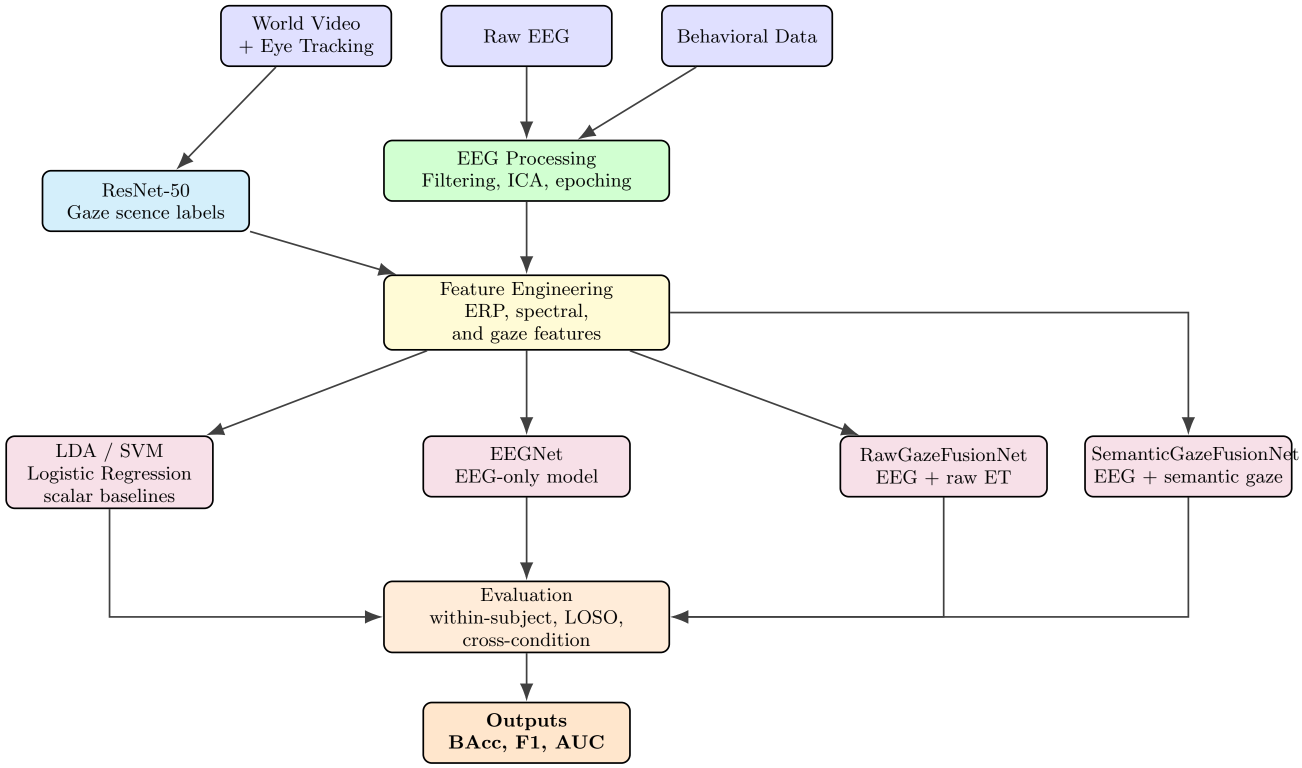

FIG. 2. Analysis pipeline for EEG preprocessing, computer-vision-based gaze classification, feature extraction, model comparison, and evaluation.

TABLE I. Distribution of hand-labeled gaze-contingent crops used to train the ResNet-50 scene classifier ($N = 8{,}570$).

Category	**N**	**%**
Trail/ground	6,576	76.7
Vegetation	1,468	17.1
People	202	2.4
Other	129	1.5
Sky	86	1.0
Water	85	1.0
Ocean	24	0.3

TABLE II. Category distribution of ResNet-50 classified fixations across the full gaze-contingent crop set ($N = 131{,}511$).

Category	**N**	**%**
Trail/ground	91,314	69.4
Vegetation	29,471	22.4
People	3,661	2.8
Other	3,528	2.7
Sky	1,829	1.4
Water	1,708	1.3
Total	131,511	100.0

was hand-labeled by five independent human labelers, with strong inter-rater reliability (Fleiss' $\kappa = 95.3\%$). A fine-tuned ResNet-50 model was trained on the labeled pixel crops using a 70/15/15 train/validation/test split. Because the labeled data were highly imbalanced, with 76.7% of samples labeled as trail/ground (Table I), focal loss and weighted sampling were used during training. The resulting classifier showed strong per-class performance, achieving 92.9% overall accuracy, (Figure 4), and its predictions across the full crop set are summarized in Table II. The temporal distribution of the resulting ResNet-50 gaze classifications across the task epoch is shown in Figure 5. The ocean label was merged into the water category at load time using a label-merge map, resulting in an effective water count of 109 samples (1.3%) for training.

D. Feature Engineering

From the preprocessed and fused EEG epochs (-200 to 800 ms relative to stimulus onset), a set of theoretically motivated scalar features was extracted per trial to serve both as interpretable scalar baselines and as inputs to classical classifiers alongside the deep learning pipeline. ERP components were quantified as mean amplitude (µV) averaged across a priori channel clusters within canonical time windows: the P300 was extracted at Pz and across a parieto-occipital cluster (300–500 ms); the N200 at a frontocentral cluster (Fz, FC1, FC2; 180–260 ms); and the NoGo-P3, a frontal positivity associated with successful response inhibition, at Fz and Cz (300–500 ms). Oscillatory power features were computed

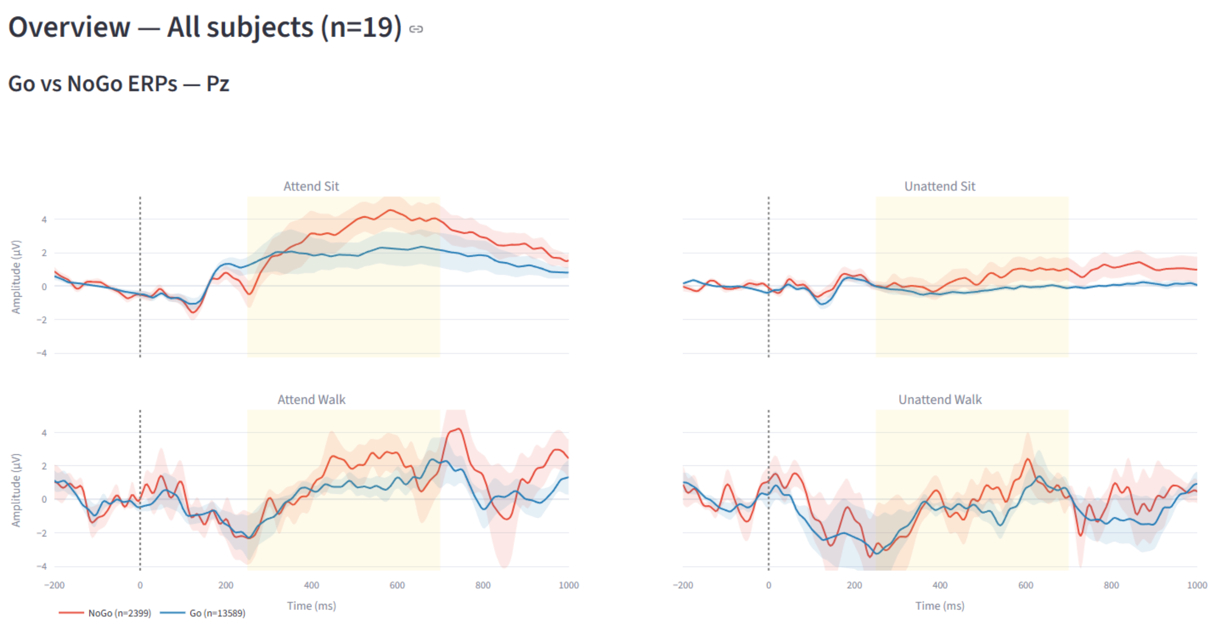

FIG. 3. P3 at electrode Pz after preprocessing (Go vs. NoGo). Subject 12 was excluded from the group ERP visualization due to absent ERP components across all conditions, attributed to poor electrode contact (FC6, T8 non-functional). Subject 12's data is still included in model training, since the EEG data remains usable as input even without clean ERPs.

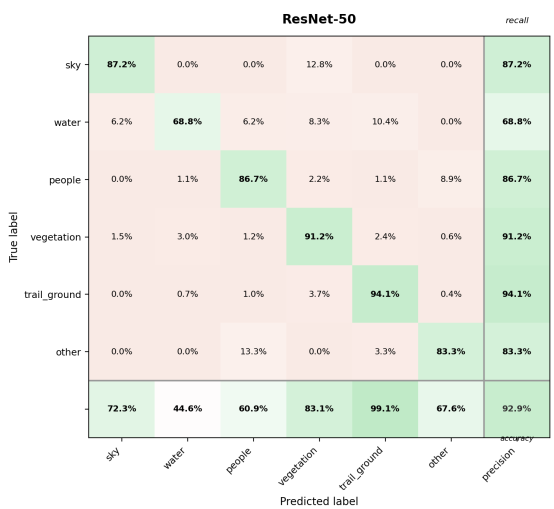

FIG. 4. Confusion matrix for the ResNet-50 scene classifier on the held-out labeled data.

using Welch's method on the trial epoch: frontal theta (4–8 Hz, Fz/F3/F4, 0–500 ms), central beta (13–30 Hz, Cz/C3/C4, 200–600 ms), and alpha power (8–12 Hz) in a pre-stimulus baseline window (-200–0 ms) separately for frontal, parietal, and occipital clusters. These ten scalar features were used as inputs to logistic regression, SVM (RBF kernel), and LDA classifiers, each with balanced class weighting, providing interpretable baselines against which deep learning performance was compared.

TABLE III. EEGNet model parameters.

Layer	Parameters
Temporal convolution	$F_1 = 8$
Depthwise convolution	$D = 2$
Feature maps	$F_1 \times D = 16$
Pointwise convolution	$F_2 = 16$
Dropout	$p = 0.5$

E. Classification Methods

The classification task represents each trial as an epoched EEG matrix $X \in \mathbb{R}^{C\times T}$, where $C = 32$ channels and $T = 251$ time points sampled at 250 Hz. The model outputs a binary prediction $\hat{y} \in \{0, 1\}$, distinguishing correct rejections (CRs) from commission errors (CEs) on NoGo trials. Similarly, for the Go vs. NoGo comparison, the model distinguishes standard Go-response trials from NoGo inhibition trials.

In this experiment, we build upon EEGNet as a compact CNN designed specifically for EEG-based BCI classification. EEGNet is advantageous for our use case because it requires minimal preprocessing and generalizes well across multiple BCI paradigms, so this experiment uses it as the backbone of the multimodal model (Lawhern et al. 2018). Its layer configuration is summarized in Table III.

Because the raw eye-tracking (ET) signal is available as a continuous time series, RawGazeFusionNet fuses it directly with EEG without any pre-labeled gaze cate-

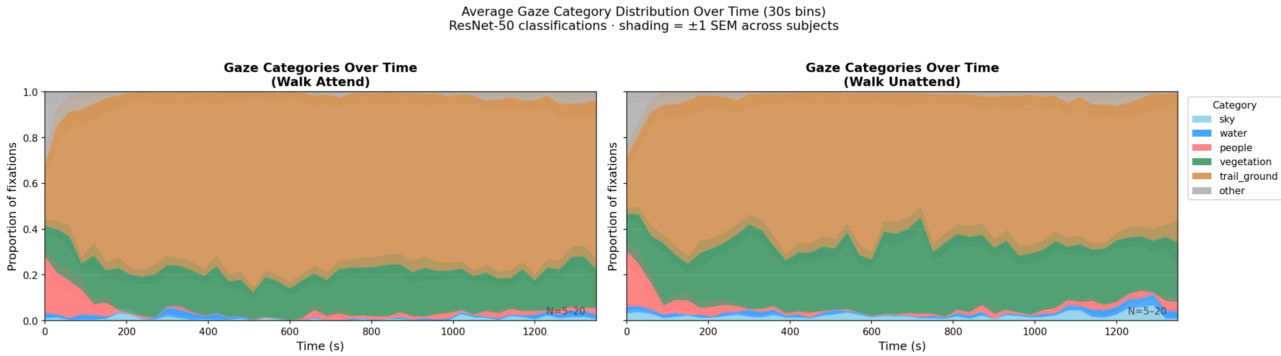

FIG. 5. Average gaze-category timeline across trials. The figure summarizes how ResNet-50 gaze classifications evolve over time relative to the task epoch.

gories.

The two-branch architecture includes an EEG branch, the EEGNet described above, and an ET branch consisting of a small 1D CNN applied to the gaze time series $g \in \mathbb{R}^{5\times T}$, where the five channels represent gaze position, pupil size, and related eye-tracking signals. Fusion is performed by concatenating the two embeddings, z_{EEG} and z_{ET}, and passing the combined representation through a fully connected layer followed by a softmax output. This late-fusion approach allows each branch to learn modality-specific features before combining EEG and eye-tracking information, consistent with prior BCI work [Ding et al., 2019].

SemanticGazeFusionNet differs from RawGazeFusionNet by using semantic pre-stimulus gaze context rather than the raw ET time series alone. The input is a sequence of fixation category labels $s_{1:L} \in \mathbb{Z}^L$, where $L < 20$ fixations are drawn from a 2000 ms pre-stimulus window and each label is produced by the ResNet scene classifier described above. Each fixation label s_t is passed through a learned embedding layer $e(s_t) \in \mathbb{R}^{32}$. The embedded sequence is then processed with a bidirectional LSTM, $h_t = \mathrm{BiLSTM}(e(s_t), h_{t-1})$. The final gaze representation, z_{gaze}, concatenates the forward and backward hidden states, allowing the model to capture temporal context across the fixation sequence. The model is trained with binary cross-entropy loss, $\mathcal{L} = -\mathbb{E}[\log p(y \mid x)]$. As non-deep-learning baselines, the hand-crafted ERP and spectral features described in the Feature Engineering section (P300, N200, and NoGo-P3 amplitudes together with frontal theta, central beta, and frontal/parietal/occipital alpha power) are fed into logistic regression, SVM, and LDA classifiers.

V. EXPERIMENT SETUP

Given the limited number of NoGo trials, a stratified 3-fold cross-validation scheme was applied per subject to evaluate within-subject performance on the correct rejection versus commission error classification problem, ensuring balanced class proportions across folds; results were then averaged across subjects. Channel-wise z-score normalisation was computed on the training folds and applied to the held-out fold to prevent data leakage. To evaluate cross-subject generalisation, a leave-one-subject-out (LOSO) cross-validation was additionally run, where the model is trained on all subjects except one and tested on the held-out subject, reflecting the more demanding and ecologically relevant scenario of deploying a BCI on a new user.

Four models were evaluated to systematically assess the contribution of each pipeline component. Scalar baselines consisted of Logistic Regression, SVM (RBF kernel), and LDA classifiers trained on hand-crafted ERP and spectral features, including: P300, N200, NoGo-P3 amplitude, frontal theta, central beta, and frontal/parietal/occipital alpha power, serving as the non-deep-learning comparison floor. EEGNet was evaluated as the unimodal deep learning baseline, operating on EEG alone with no eye-tracking input. RawGazeFusionNet extended EEGNet by fusing the raw ET time series through a parallel 1D CNN branch via late fusion. Finally, SemanticGazeFusionNet replaced the raw ET branch with a bidirectional LSTM over the pre-stimulus fixation category sequence, testing whether the semantic content of where participants looked before the stimulus provided additional discriminative signal. To assess whether SemanticGazeFusionNet meaningfully improved over EEGNet, a Wilcoxon signed-rank test was applied across per-fold balanced accuracy scores for these two models.

All deep learning models were trained using the AdamW optimiser (lr = 0.001, weight decay = 0.01) with a cosine annealing learning rate schedule, a batch size of 32, and a maximum of 100 epochs. Early stopping with a patience of 10 epochs was applied, halting training when validation loss failed to improve. Dropout of 0.25 was used in EEGNet consistent with the original specification; the gaze encoder and fusion head in SemanticGazeFusionNet used a dropout of 0.3. Given the class imbalance between correct rejections and commission errors inherent to the Go/NoGo paradigm, class-weighted cross-entropy loss was used, with weights inversely proportional to class frequency, to prevent the majority class from dominating training. All models were trained on an NVIDIA RTX 4090.

VI. RESULTS & DISCUSSION

A. Correct rejection vs Commission Error in NoGo trials

Full metrics for this task are reported in Table IV, with significance tests in Table V; here we focus on the key results. On the walk-attend condition, EEGNet was the strongest model under cross-subject (LOSO) evaluation, reaching 61.03% ± 1.44% balanced accuracy (64.66% AUC, 70.67% F1). It performed reliably above chance ($W = 190$, $p < .001$) and significantly better than every scalar baseline ($p < .001$), none of which exceeded chance. Adding gaze did not help, neither RawGazeFusionNet nor SemanticGazeFusionNet differed significantly from EEGNet ($p = .332$ and $p = .798$). For inhibitory-control classification, then, gaze, whether raw or semantically labeled, carries little discriminative signal beyond EEG.

Three further patterns stand out (Table IV). First, within-subject accuracy (58.53%) was slightly below LOSO; because each within-subject model was trained on only about 150 trials, this likely reflects limited training data, a consequence of the small number of failed-inhibition trials per subject, rather than genuinely easier cross-subject transfer. Second, performance was higher when sitting than walking (62.84% vs. 58.92% within-subject), consistent with locomotion adding noise and competing cognitive load [Pizzamiglio et al., 2017; Patelaki et al., 2022; Peskar et al., 2023]. Third, cross-condition transfer was asymmetric: walk-to-sit (61.52%) exceeded sit-to-walk (57.24%), suggesting that the greater variability of walking data yields more transferable representations. Few-shot calibration (Table VI) did not improve on the unadapted ($k = 0$) model, plateauing near 57–58% balanced accuracy.

B. Go vs NoGo Classification

The Go vs. NoGo task showed the same qualitative pattern (Table VII; significance in Table VIII). EEGNet was again best, reliably above chance ($p < .001$), significantly better than all scalar baselines ($p < .001$), and not significantly improved by gaze fusion ($p = .275$ and $p = .169$). Balanced accuracy was 61.09% ± 1.74% within-subject and 58.42% ± 1.08% under LOSO. The low precision and F1 across all models (EEGNet F1 = 33.14%) reflect the strong class imbalance of the go/nogo paradigm, in which NoGo trials are rare; balanced accuracy is therefore the more meaningful metric here.

Locomotion mattered more for this task: within-subject accuracy fell from 70.70% (sitting) to 61.46% (walking), a 9.24-point gap versus 3.92 points for the inhibition task, likely because separating Go from NoGo relies on ERP morphology that motion distorts more readily. Cross-condition transfer was again asymmetric, with walk-to-sit (63.63%) exceeding sit-to-walk (56.69%).

a. Summary of Key Findings. Across both tasks, four results stand out. (1) EEGNet substantially outperforms the traditional baselines, reaching roughly 58–61% cross-subject balanced accuracy under real-world walking, while the scalar models stay near chance. (2) Contrary to our gaze-fusion hypothesis, neither raw nor semantic gaze fusion significantly improved on EEG alone in either task. (3) Walking reliably degraded performance relative to sitting (about 3–5 points for inhibition and 7–9 points for Go/NoGo), consistent with locomotion-induced artifacts and dual-task load. (4) Cross-condition transfer was asymmetric, with walking-trained models generalizing better to sitting than the reverse. Together, these results indicate that subject variability and motion artifacts, rather than the choice of input modality, are the dominant obstacles for mobile EEG BCIs, pointing toward domain adaptation or online learning as the more promising route to real-world robustness.

TABLE IV. **Correct inhibition vs failed inhibition classification.** Summary results for EEG-based correct rejection (CR) versus false alarm (FA) classification across evaluation settings. Average percentage metrics for Balanced Accuracy (BAcc), AUC-ROC (AUC), and F1 score are reported as mean ± SE where available; single-subject evaluations report point estimates only. **Bold** = best per column within each part. $N = 19$ subjects for walking scope; $N = 20$ for attend-all scope.

	Within-Subject			**LOSO**		
Method	BAcc(%)	AUC	F1	BAcc(%)	AUC	F1
Part A: Walk-Attend condition						
EEGNet	**58.53 ± 1.52**	61.15 ± 1.86	65.06 ± 2.41	**61.03 ± 1.44**	**64.66 ± 1.73**	70.67 ± 1.52
RawGazeFusionNet	55.57 ± 1.55	59.53 ± 1.97	65.97 ± 2.55	58.81 ± 0.84	63.12 ± 1.24	**71.06 ± 1.62**
SemanticGazeFusionNet	58.33 ± 1.56	**61.56 ± 1.80**	**67.29 ± 3.00**	59.68 ± 1.56	63.56 ± 2.10	69.25 ± 1.66
Logistic Regression	51.41 ± 1.22	52.11 ± 1.57	65.66 ± 2.02	51.23 ± 0.96	52.12 ± 1.53	45.90 ± 2.86
SVM	50.87 ± 1.37	51.01 ± 1.56	66.29 ± 2.62	51.34 ± 0.89	53.84 ± 1.21	48.36 ± 4.00
LDA	51.68 ± 1.18	51.66 ± 1.49	62.26 ± 1.76	50.39 ± 1.02	52.00 ± 1.70	45.56 ± 3.29
Part B: All Attend conditions (Sit & Walk)						
EEGNet	**60.37 ± 1.87**	**63.91 ± 2.51**	**70.71 ± 2.52**	**61.49 ± 1.43**	**66.04 ± 1.86**	**70.45 ± 2.43**
Logistic Regression	53.17 ± 0.92	54.15 ± 1.17	64.86 ± 2.28	49.88 ± 0.80	50.67 ± 0.88	45.21 ± 2.82
SVM	52.88 ± 1.24	51.94 ± 1.40	66.73 ± 2.69	49.57 ± 0.71	51.09 ± 0.85	37.54 ± 4.91
LDA	51.59 ± 0.89	52.00 ± 1.15	62.48 ± 2.05	50.01 ± 0.70	49.91 ± 1.09	43.90 ± 3.23
Part B: Per-condition breakdown (within-subject, point estimates)						
	sit_attend			*walk_attend*		
EEGNet	**62.84**	**71.18**	**76.89**	**58.92**	**63.25**	**69.06**
Logistic Regression	54.59	56.51	70.29	51.90	53.12	63.34
SVM	55.12	60.35	73.41	51.86	65.02	66.54
LDA	52.71	54.19	66.94	49.67	51.08	60.92
Part C: Cross-condition generalization (point estimates)						
	sit → walk			*walk → sit*		
EEGNet	**57.24**	**61.30**	**63.30**	**61.52**	**68.15**	**79.74**
Logistic Regression	49.12	49.23	63.14	48.71	48.43	34.36
SVM	49.99	50.08	70.94	49.37	50.20	42.39
LDA	49.88	49.16	61.98	46.91	47.42	42.80

TABLE V. Wilcoxon signed-rank tests on per-subject AUC values (one-tailed vs. chance; two-tailed for pairwise model comparisons).

	Part A: Walking ($n = 19$)		**Part B: Attend-all** ($n = 20$)	
Comparison	W	p	W	p
vs. chance (one-tailed)				
EEGNet	190	<.001	209	<.001
RawGazeFusionNet	190	<.001		—
SemanticGazeFusionNet	185	<.001		—
Scalar baselines (all)		$p \geq .091$		$p \geq .108$
vs. EEGNet (two-tailed)				
LogisticRegression	6	<.001	—	<.001
SVM	10	<.001	—	<.001
LDA	8	<.001	—	<.001
RawGazeFusionNet	70	.332 ns		—
SemanticGazeFusionNet	88	.798 ns		—

— = not tested in this condition; ns = non-significant.

TABLE VI. **Few-shot subject adaptation for correct inhibition vs failed inhibition.** Part D results for EEG classification on the CR versus FA task. Models are evaluated with $k \in \{0, 5, 10, 20\}$ calibration trials per subject. The $k = 0$ row corresponds to pure leave-one-subject-out evaluation without adaptation. Average percentage metrics are reported as mean ± SE. **Bold** = best per column within each scope.

Method	k	BAcc(%)	AUC	F1
Scope: `walk_attend`				
EEGNet	0	57.19 ± 1.30	60.61 ± 1.68	67.81 ± 1.39
	5	57.80 ± 1.20	60.48 ± 1.59	69.62 ± 1.32
	10	57.53 ± 1.14	$\mathbf{61.33 \pm 1.56}$	68.43 ± 1.80
RawGazeFusionNet	0	$\mathbf{58.95 \pm 0.82}$	$\mathbf{61.14 \pm 1.29}$	69.14 ± 1.62
	5	57.24 ± 1.12	60.72 ± 1.41	66.45 ± 1.92
	10	57.33 ± 1.23	60.56 ± 1.50	68.61 ± 1.77
SemanticGazeFusionNet	0	57.69 ± 1.29	60.17 ± 1.70	$\mathbf{69.29 \pm 1.62}$
	5	57.20 ± 1.39	59.85 ± 1.56	66.47 ± 2.16
	10	57.24 ± 1.53	59.74 ± 1.66	69.04 ± 1.74
Scope: `(walk attend & sit attend)`				
EEGNet	0	$\mathbf{59.99 \pm 1.53}$	63.30 ± 2.00	69.57 ± 1.77
	5	59.63 ± 1.47	62.79 ± 1.96	68.45 ± 2.03
	10	59.56 ± 1.64	$\mathbf{63.74 \pm 1.99}$	68.34 ± 2.18
	20	59.91 ± 1.57	63.11 ± 1.92	$\mathbf{69.69 \pm 2.02}$
LDA	5	50.12 ± 1.20	49.92 ± 1.39	58.49 ± 3.40
	10	49.34 ± 0.79	49.61 ± 1.26	58.65 ± 2.44
	20	52.17 ± 0.86	52.19 ± 0.96	65.48 ± 1.75
SVM	5	50.34 ± 0.78	49.46 ± 1.02	69.41 ± 3.85
	10	50.35 ± 0.91	50.17 ± 1.23	55.80 ± 3.53
	20	50.77 ± 1.12	49.35 ± 1.27	58.66 ± 3.78
Logistic Regression	5	50.46 ± 0.89	50.28 ± 1.17	63.90 ± 3.11
	10	49.73 ± 1.05	49.85 ± 1.30	56.59 ± 3.38
	20	52.61 ± 0.64	52.88 ± 0.98	63.84 ± 2.06

TABLE VII. **Go vs NoGo trial-type classification.** Summary results for EEG-based Go versus NoGo classification across evaluation settings. This task distinguishes standard response trials from NoGo inhibition trials. Average percentage metrics for Balanced Accuracy (BAcc), AUC-ROC (AUC), and F1 score are reported as mean ± SE where available; single-condition evaluations report point estimates only. **Bold** = best per column within each part. $N = 19$ subjects for walking scope (one subject walking-only); $N = 20$ for attend-all scope.

	Within-Subject			**LOSO**		
Method	BAcc(%)	AUC	F1	BAcc(%)	AUC	F1
Part A: Walk Attend condition						
EEGNet	$\mathbf{61.09 \pm 1.74}$	$\mathbf{65.28 \pm 2.27}$	$\mathbf{33.14 \pm 2.04}$	58.42 ± 1.08	62.58 ± 1.42	29.22 ± 1.14
RawGazeFusionNet	58.49 ± 1.82	61.67 ± 2.25	30.16 ± 2.06	58.71 ± 1.38	62.75 ± 1.70	29.53 ± 1.40
SemanticGazeFusionNet	59.26 ± 1.83	62.85 ± 2.28	30.88 ± 1.96	$\mathbf{58.78 \pm 1.12}$	$\mathbf{62.96 \pm 1.47}$	$\mathbf{29.57 \pm 1.25}$
Logistic Regression	51.32 ± 0.74	52.29 ± 1.01	24.17 ± 0.57	50.26 ± 0.69	49.94 ± 0.88	22.75 ± 0.66
SVM	52.11 ± 0.66	51.76 ± 1.11	24.35 ± 0.61	50.14 ± 0.62	49.39 ± 0.74	24.17 ± 0.89
LDA	51.48 ± 0.68	51.85 ± 0.93	24.36 ± 0.53	51.05 ± 0.86	50.89 ± 1.15	24.26 ± 0.78
Part B: All Attend conditions (Sit & Walk)						
EEGNet	$\mathbf{58.97 \pm 0.94}$	$\mathbf{62.98 \pm 1.37}$	$\mathbf{30.36 \pm 0.98}$	$\mathbf{56.49 \pm 0.71}$	$\mathbf{59.62 \pm 0.89}$	$\mathbf{27.05 \pm 0.84}$
Logistic Regression	50.63 ± 0.57	50.72 ± 0.73	23.67 ± 0.41	51.19 ± 0.43	51.04 ± 0.53	23.69 ± 0.41
SVM	51.37 ± 0.60	51.99 ± 0.78	23.78 ± 0.60	51.31 ± 0.36	51.47 ± 0.50	24.68 ± 0.34
LDA	51.05 ± 0.45	51.28 ± 0.58	23.55 ± 0.43	50.48 ± 0.45	50.27 ± 0.56	23.95 ± 0.37
Part B: Per-condition breakdown (within-subject, point estimates)						
	sit_attend			*walk_attend*		
EEGNet	**70.70**	**78.39**	**42.19**	**61.46**	**66.73**	**32.65**
Logistic Regression	53.04	54.24	25.91	51.15	51.70	23.76
SVM	53.65	55.10	26.21	51.60	52.60	24.06
LDA	53.11	54.33	25.53	51.38	51.66	23.79
Part C: Cross-condition generalization (point estimates)						
	sit → walk			*walk → sit*		
EEGNet	**56.69**	**59.63**	**27.98**	**63.63**	**69.14**	**35.03**
Logistic Regression	50.99	51.36	23.92	50.08	50.52	22.91
SVM	51.48	51.43	24.29	51.64	45.65	26.47
LDA	50.36	50.38	23.93	51.24	52.07	19.94

TABLE VIII. Wilcoxon signed-rank tests on per-subject AUC values (one-tailed vs. chance; two-tailed for pairwise model comparisons).

	Part A: Walking ($n = 19$)		**Part B: Attend-all** ($n = 20$)	
Comparison	W	p	W	p
vs. chance (one-tailed)				
EEGNet	190	<.001	210	<.001
RawGazeFusionNet	189	<.001		—
SemanticGazeFusionNet	190	<.001		—
Scalar baselines (all)		$p > .22$		$p > .14$
vs. EEGNet (two-tailed)				
LogisticRegression	—	<.001	—	<.001
SVM	—	<.001	—	<.001
LDA	—	<.001	—	<.001
RawGazeFusionNet	67	$.275^{\text{ns}}$		—
SemanticGazeFusionNet	60	$.169^{\text{ns}}$		—

— = not tested or not reported in this condition; ns = non-significant.

VII. LIMITATIONS

Several limitations should be kept in mind when interpreting these results. First, the data were in places missing or degraded. Subject 12 lacked clean ERP components owing to poor electrode contact, and Subject 20 had to be dropped from some analyses because the behavioral triggers could not be reliably synchronized with the EEG. More broadly, many participants contributed very few failed-inhibition trials (13 subjects had fewer than roughly 50), which sharply limited the data available for the correct-rejection versus commission-error problem. Combined with a relatively small and highly imbalanced dataset, both across fixation categories (trail/ground dominates) and across Go/NoGo trial types, this constrains the classification performance achievable in principle. Substantial inter-subject variability and motion-related noise further limit cross-subject generalization, as reflected in the gap between within-subject and LOSO performance.

Second, our null result for gaze fusion should be read narrowly. Under the present conditions adding gaze did not improve classification, but this does not imply that gaze is uninformative in general. Participants were largely free-viewing while walking rather than using their eyes in a goal-directed way tied to the task, so there was little structured gaze behavior for the model to exploit. A different task, activity, or set of lighting conditions, particularly one in which gaze location is itself task-relevant, could produce a much stronger interaction between the two modalities. Relatedly, awareness of the recording setup may have led participants to behave less naturally and look around less than they otherwise would, further reducing the informativeness of the gaze stream.

Third, while these findings are encouraging as basic science, the performance is not yet sufficient for a deployed BCI. Even with research-grade equipment, the signal-to-noise ratio of scalp EEG recorded during movement is low, and balanced accuracies in the high 50s to low 60s fall well short of what reliable real-world control would require. This raises broader questions about the efficacy of non-invasive BCIs for demanding, mobile use cases. Our analyses were also entirely offline; a deployed system would operate at best in near-real-time and would have to contend with the processing latency of the pipeline as well as the lag and synchronization between the EEG and eye-tracking streams, the very kind of timing issue that forced us to exclude Subject 20.

Despite these constraints, the present work is best viewed as a proof of concept, it shows that a compact deep-learning model can recover above-chance inhibitory-control and Go/NoGo signals from mobile EEG recorded during real-world walking. Many of the limitations above are addressable with better technology and more data. Higher-density or, ultimately, implanted electrodes would provide substantially higher signal-to-noise and spatial resolution [Schwartz et al., 2006]; improved artifact handling, much larger and better-balanced datasets, and online or adaptive training would directly target the data and generalization bottlenecks observed here. We see these as the natural next steps toward turning this proof of concept into a usable mobile BCI.

CONCLUSION

In this paper we presented an outdoor, real-world walking dataset for developing BCI classification models, and evaluated it against our three hypotheses. Contrary to the first hypothesis, adding vision as a modality, whether raw gaze (RawGazeFusionNet) or semantic gaze context (SemanticGazeFusionNet), provided no meaningful boost over EEG alone for inhibitory-control classification. The remaining two hypotheses were supported: walking reliably hurt performance, consistent with the literature on physical activity and dual-task cognition, and cross-condition transfer was asymmetric, with walking-trained models generalizing better to sitting than the reverse. Overall, EEGNet proved a viable backbone for mobile BCI under real-world conditions, reaching approximately 58–61% balanced accuracy across subjects. These results were obtained despite real constraints on data quantity, balance, and signal quality, which we examine in the Limitations along with the steps most likely to address them. We highlight one avenue in particular: although gaze fusion did not help here, it may yet prove valuable in settings where eye movements are tightly coupled to the task, an opportunity worth pursuing as mobile BCIs move toward real-world deployment.

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